LAB 14

# Pairs Stage: a Hedged, Market-Neutral Statistical-Arbitrage Sleeve

| Item | Detail |
|---|---|
| Topic | Topic 3: Machine Learning Models, Statistical Modelling and A/B Testing for Trading Signals |
| Pipeline stage | Pairs / stat-arb |
| TSC reference | A4, K6 |
| Duration | 20 min |
| Folder | labs_fullcode/Lab14-pairs-trading/ |
| Files | Lab14-pairs-trading/pairs_stage.py, pipeline/pairs.py, data/KO.csv, data/PEP.csv, data/BAC.csv, Lab14-pairs-trading/solution/pairs.py |
| Prompts | Lab14_Prompts.pdf in this folder (step -> prompt, ready to copy) |
| Mode | Runs ONLINE with your Alpaca paper keys (.env), or OFFLINE on the bundled data and the paper-broker simulator. Paper trading only. |

*Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.*

## Scenario

Lion City Capital's equity desk wants a sleeve whose returns do not depend on the direction of the market. The head of trading asks for the classic statistical-arbitrage hedge as a pipeline stage: hold one stock long and an economically linked stock short in hedge-ratio proportion, trade the spread when it stretches, and output per-leg target weights. Candidates: KO/PEP, JPM/BAC and XLF/JPM.

## Objective

Build pipeline/pairs.py: screen pairs with the Engle-Granger cointegration test, estimate the hedge ratio, trade the spread z-score out of sample with next-bar execution and costs on both legs, confirm market neutrality and output leg targets.

## Before You Start

- Open a terminal in the labs_fullcode folder, activate your virtual environment and open your AI coding assistant there (set-up below). The complete, working code is already in this folder.
- Record every result you are asked to check - the same evidence is asked for in the Practical Performance assessment.

## How to Run the Python Scripts

*Everything runs from the labs_fullcode folder, ONLINE with your Alpaca paper keys or OFFLINE on the bundled data. Do the first four rows once; after that each lab is one command.*

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| Step | Command / file |
|---|---|
| Open a terminal in the labs_fullcode folder | cd labs_fullcode |
| Create and activate a virtual environment (once) | python3 -m venv .venv<br>source .venv/bin/activate (Windows: .venv\Scripts\activate) |
| Install the libraries (once) | pip install -r requirements.txt |
| Optional: your Alpaca paper keys (leave blank to run OFFLINE) | cp .env.example .env (Windows: copy .env.example .env)<br>then paste APCA_API_KEY_ID and APCA_API_SECRET_KEY into .env |
| Run Lab 14 | python3 Lab14-pairs-trading/pairs_stage.py<br>(same as: python -m pipeline pairs) |
| The complete code | pipeline/pairs.py<br>(reference copy: Lab14-pairs-trading/solution/) |
| Outputs | _outputs/ (charts, CSV files and reports written by the stage) |

## Lab Steps - In Python with Your AI Assistant

*AI writes the code; you own the algorithm. Paste each prompt into your AI coding assistant opened in the labs_fullcode folder (the same prompts, ready to copy, are in Lab14-pairs-trading/Lab14_Prompts.pdf and Prompt_Library.html), run the stage and check the result. The complete, working code is already provided - see 'How to Run the Python Scripts' above. The Full Code Practical Performance uses this code path.*

### Step 1: Read the method (K6)

Read the docstring of Lab14-pairs-trading/solution/pairs.py: the formation period 2018-2022 is used only to estimate the hedge ratio and test cointegration; the trading period 2023-2025 is out of sample with the hedge ratio held fixed.

*Check: You can explain why the hedge ratio must be frozen before the trading period starts.*

### Step 2: Build the cointegration tools (K6)

Ask your assistant for the statistics part of pipeline/pairs.py.

| AI prompt - paste into your AI coding assistant |
|---|
| Create labs_fullcode/pipeline/pairs.py with numpy-only statistics: ols(y, X) returning coefficients, standard errors and residuals; adf_t(series, lags=1) - the ADF t-statistic of g in d e_t = g e_(t-1) + phi d e_(t-1); engle_granger(log_a, log_b) - step 1 OLS log A = a + beta log B, step 2 adf_t on the residual - returning (alpha, beta, adf_t); half_life(spread) from an AR(1) fit of spread changes; EG_CRIT = {'1%': -3.90, '5%': -3.34, '10%': -3.04} (MacKinnon, two variables with constant). FORMATION = ('2018-01-01', '2022-12-31'), TRADING = ('2023-01-01', '2025-12-31'). Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow. |

*Check: engle_granger on log KO and log PEP over the formation period returns beta about 0.756 and ADF t about -3.36.*

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### Step 3: Build the spread trading rules and leg targets (A4)

Ask your assistant for trade_pair(), leg_targets(), run() and main().

| AI prompt - paste into your AI coding assistant |
| --- |
| In labs_fullcode/pipeline/pairs.py add trade_pair(closes, a, b, alpha, beta, entry, exit_, window, cost_bps): spread = log A - beta log B - alpha; z = (spread - rolling mean) / rolling std over window; state machine: short spread (-1: short A, long B) when z > +entry, long spread (+1) when z < -entry, exit when \|z\| < exit_; held = state.shift(1) (decided at close t, held over t+1); gross = held x w x (rA - beta rB) with w = 1 / (1 + \|beta\|); costs = \|change in held\| x w x (1 + \|beta\|) x cost_bps per leg per side; return spread, z, held, state, gross, net. leg_targets(state, a, b, beta) -> {a: state x w, b: -state x beta x w}. run(pair=('KO', 'PEP'), entry=2.0, exit_=0.5, window=60, cost_bps=5.0) for `python -m pipeline pairs --pair --entry --exit --window --cost-bps`: screen config universe.pairs (correlation of returns, beta, ADF t, cointegrated at 5%?), print the chosen pair's hedge ratio and half-life (warn if not cointegrated), trade 2023-2025 using 120 days of history for the z warm-up, print gross, net and SPY return, CAGR, volatility, Sharpe and max drawdown, trades, time in market, beta and correlation to SPY, the latest leg targets, and a note that short legs need a margin account while the automation stages are long only. Save chart Lab14_pairs_trading.png (normalised prices, z with bands and open positions, equity vs SPY). Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow. |

*Check: python -m pipeline pairs --help shows --pair, --entry, --exit, --window and --cost-bps.*

### Step 4: Run the pair screen (A4, K6)

Run: python -m pipeline pairs (same as: python3 Lab14-pairs-trading/pairs_stage.py). Compare correlation with the ADF t-statistic for each pair.

*Check: KO/PEP: correlation 0.75, beta 0.76, ADF t -3.36 - cointegrated at 5% (critical -3.34). JPM/BAC 0.93, ADF -2.08 and XLF/JPM 0.94, ADF -1.89 - highly correlated but NOT cointegrated.*

### Step 5: Interpret the hedge ratio (K6)

Read section [2]. Explain in one sentence why the hedge ratio, not a 1:1 position, makes the spread stationary.

*Check: KO/PEP hedge ratio 0.756 (short 0.76 of PEP per 1 of KO), half-life 39 days.*

### Step 6: Evaluate the out-of-sample result (A4)

Read section [3] and [4]: trades, time in market, beta to SPY and the latest leg targets.

*Check: 2023-2025: 13 trades, 59% time in market, gross -9.4%, net -10.6% (Sharpe -0.60), beta to SPY +0.02, correlation +0.04, versus SPY +86.3%; latest z +0.65 -> spread position 0 -> leg targets KO 0.0, PEP 0.0. The hedge worked (beta near 0), but the spread did not revert profitably after 2022.*

### Step 7: Test a failed pair and other thresholds (A4, K6)

Run: python -m pipeline pairs --pair JPM BAC --entry 1.5 --exit 0. Explain why trading a pair that failed the cointegration test is a statistical error and why trying many pairs and thresholds is data snooping.

*Check: JPM/BAC: 'WARNING: NOT cointegrated' (ADF -2.08, half-life 93 days); 1 trade held 100% of the time, net -21.9%, beta to SPY -0.03.*

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### Step 8: Study the chart (A4)

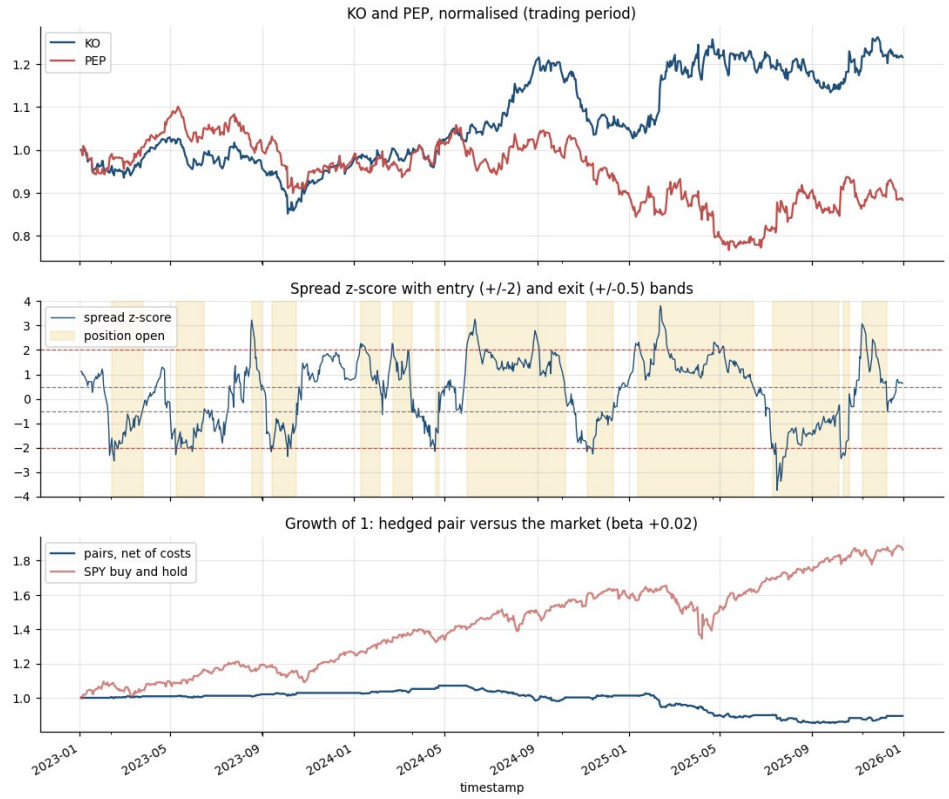
Open _outputs/Lab14_pairs_trading.png: normalised prices, the z-score with entry and exit bands and open positions shaded, and the equity curve against SPY.

| AI prompt - paste into your AI coding assistant |
| --- |
| Looking at labs_fullcode/_outputs/Lab14_pairs_trading.png and the stage output, explain to the head of trading how the desk should re-test cointegration on a rolling basis and size each pair small before this sleeve could ever be traded, and why it stays a paper sleeve while the automation stages are long only. |

*Check: A rolling re-test plan (for example every quarter on the last 3 years) and a position-size limit per pair.*

## Expected Output (Full Code)

python -m pipeline pairs: screening KO/PEP ADF t -3.36 (cointegrated at 5%), JPM/BAC -2.08 and XLF/JPM -1.89 (not); KO/PEP hedge ratio 0.756, half-life 39 days; out of sample 2023-2025: 13 trades, 59% time in market, gross -9.4%, net -10.6% (Sharpe -0.60), beta to SPY +0.02, versus SPY +86.3%; leg targets KO 0.0 / PEP 0.0 at 2025-12-31; chart _outputs/Lab14_pairs_trading.png.




*Lab 14 output (labs_fullcode/_outputs/Lab14_pairs_trading.png), offline mode on the bundled data.*

## Debrief

Pairs trading is a hedging algorithm: the short leg cancels most of the market exposure of the long leg, which is why the beta to SPY is near zero (A4). The statistical model is cointegration, not correlation (K6). A pair that is cointegrated in one period can stop reverting in the next, so professional desks re-test cointegration on a rolling basis and size each pair small.

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## Troubleshooting

| Symptom | What to do |
|---|---|
| KeyError for KO, PEP or BAC | The bundled data folder is out of date; copy data/KO.csv, data/PEP.csv and data/BAC.csv from the course lab folder. |
| ADF t differs slightly online | Alpaca's IEX feed and the bundled Yahoo data differ a little; use the bundled data for the class comparison. |
| All pairs show 'no' | Check the formation period is 2018-2022 and that log prices are used. |

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